

From Raw Wearable Signals to an Analysis-Ready HRV Pipeline: Dataset Characterization, Missingness Handling, Smoothing Methodology, and Candidate Change-Point Detection Methods

Jatin Dangi¹ and Aarush Ram Anandh¹

¹Koita Centre for Digital Health (KCDH)

June 3, 2026

Abstract

This manuscript documents the construction of an analysis-ready Heart Rate Variability (HRV) pipeline built on multimodal wearable recordings supplied by our study partner. We describe the dataset, characterize the HRV signal across ten properties (H1–H10) that jointly inform every downstream methodological choice, formalize a principled approach to missingness, and develop a smoothing stage based on a structural particle smoother with harmonic components. We additionally survey and motivate a set of candidate change-point detection (CPD) methods that build on the smoothed signal. The present paper concludes at the point of presenting and motivating these candidate CPD methods; the implementation, experimentation, and downstream evaluation of CPD are reserved for subsequent work.

Contents

1	Introduction	3
2	Dataset and Signal Characteristics	4
2.1	Starting Point: Data Handover	4
2.2	HRV Signal Characteristics (H1–H10)	5
3	Data Handling and Preprocessing	6
3.1	Per-Patient Ingestion and HRV Isolation	6
3.2	Exploratory Data Analysis	6
3.3	Characterizing Missingness	6
3.4	The 180-Minute Chunking Rule	6
4	Handling Missingness	7
4.1	Why Standard Imputation Fails Here	7
4.2	Where We Do and Do Not Impute	8
5	Smoothing	8
5.1	Why Smoothing in the First Place	8
5.2	Cross-Disciplinary Survey and Scoring System	8
5.3	Selection of the Particle Filter	8

6	Structural Particle Smoother with Harmonic Components	8
6.1	Overview and Motivation	8
6.2	State Vector and Initial Distribution	9
6.3	State Transition Model	10
6.3.1	Adaptive Time Increment	10
6.3.2	Local Linear Trend Transition	10
6.3.3	Trigonometric Harmonic Transition	10
6.3.4	Observable Equation	11
6.4	Observation Model and Missing Data	11
6.4.1	Student- t Likelihood	11
6.4.2	Missing Observation Handling	12
6.5	Data-Driven Parameter Estimation	12
6.6	Sequential Monte Carlo Forward Filter	13
6.7	Forward-Filter Backward-Sampler	13
6.8	Assumptions of the Smoothing Stage	14
7	Change-Point Detection	14
8	Candidate Change-Point Detection Methods	15
9	Mathematical Formulation of Selected CPD Methods	16
9.1	Bayesian Online Change-Point Detection (BOCPD)	16
9.2	Kalman Filter for Change-Point Detection	16
9.3	Kernel Change-Point Detection (KCP)	16
9.4	Hidden Markov Models (HMM)	16
9.5	Consolidated Overlay	17
A	Smoothing Method Survey	19

1 Introduction

This work documents the pipeline construction phase of an HRV-based physiological monitoring study. The objective is not to make a clinical prediction; rather, it is to produce an analysis-ready signal whose properties are respected at every stage and whose downstream use for change-point detection is well-motivated. Figure 1 provides a complete overview of the data pipeline, from the incoming multimodal patient data through HRV processing (missingness handling, smoothing, change-point detection, annotation) and the overlay and combinatorics step that feeds the downstream machine learning model.

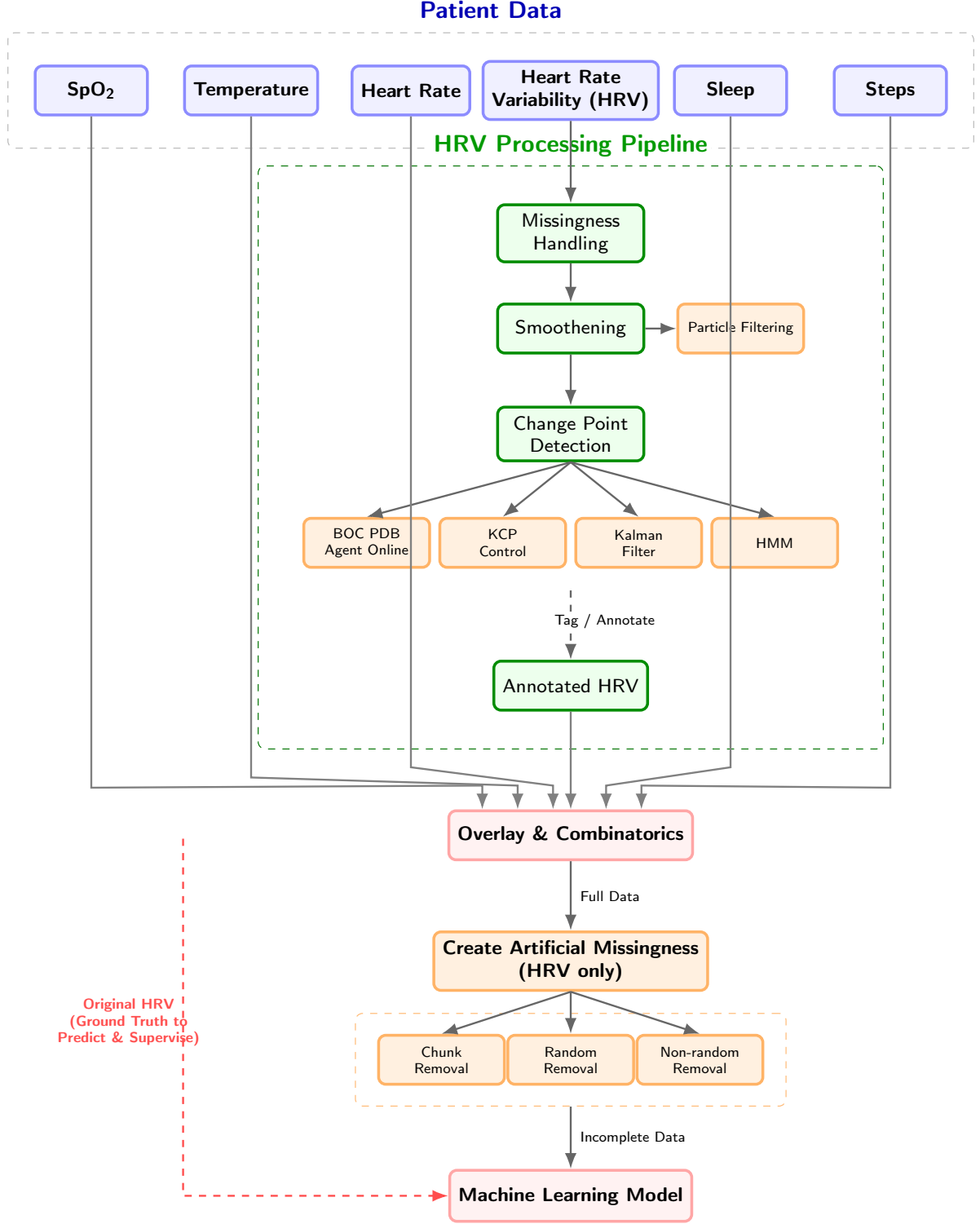


Figure 1: Complete data pipeline from incoming patient data through HRV processing, change-point detection, overlay and combinatorics, and downstream machine learning model.

2 Dataset and Signal Characteristics

2.1 Starting Point: Data Handover

Our contribution begins after data acquisition. Raw multimodal recordings provided by our study partner, GOQII including HRV (RR intervals), heart rate, SpO₂, skin temperature, step

count, and sleep stages were received in their device-native format. Before designing the pipeline, we first examined the data frequency, quality, and related attributes summarized in Table 1. We then characterized the HRV signal across ten properties that jointly informed every downstream methodological choice, as presented in Table 2.

Table 1: Dataset summary: signal modalities, trigger mechanism, cadence, and output format.

Metrics		Trigger Mechanism	Data Nature	Frequency	Duration	Output mat	For-	File Type
HRV		Automatic	Continuous	Every 10 minute	10 min	Single reading (e.g., 87)	HRV	.csv
Blood Pressure	Pres-	User-Initiated (via button/app)	Discrete	On-demand	~60 seconds	Single BP reading (e.g., 120/80 mmHg)		.csv
Sleep		Auto-detected (movement & HR analysis)	Continuous (captured) → Discrete	Throughout sleep duration	Full night	Sleep stages + total sleep duration		.csv
Heart Rate		Automatic Continuous	Continuous	Every 5 minutes	Real-time sampling	BPM values across the day		.csv
ECG		User-Initiated (via button/app)	Discrete	On-demand	~44 seconds	ECG waveform with rhythm interpretation		.csv
SpO ₂		User-Initiated (via button/app)	Discrete	On-demand	30–40 seconds	% Saturation (e.g., 96%)		.csv
Body Temperature	Temper-	Automatic Continuous	Continuous	Every 5–15 minutes	Real-time	Averaged skin/-body temperature		.csv
Steps		Automatic via motion sensors	Continuous detection → Discrete	Real-time	Real-time	Cumulative daily step count		.csv

2.2 HRV Signal Characteristics (H1–H10)

Table 2: HRV signal characteristics (H1–H10) used as the design specification for downstream choices.

Code	Natural Language	Mathematical form
H1	Bounded positive support	$X \in (0, X_{\max}]$; RR 200–1200 ms; RMSSD ≥ 0
H2	Bimodal distribution	≥ 2 modes; mixture of underlying processes (likely day vs night HRV)
H3	1/f spectrum, long memory	$S(f) \propto 1/f^\beta$; ACF decays as power law (slowly)
H4	Non-stationary	$\mu(t), \sigma^2(t), S(f, t)$ all time-varying
H5	Circadian rhythm	Deterministic ~ 24 h periodicity
H6	Heavy-tailed contamination	Motion bursts + ectopic point outliers; non-Gaussian noise
H7	Non-uniform sampling	Irregular Δt between observations
H8	Mixed missingness	Random (motion artifacts) + non-random (off-wrist, charging)
H9	Online (causal) operation	Estimator at t uses only past observations
H10	Between-patient variability	$\sigma_{\text{between}}^2 \gg \sigma_{\text{within}}^2$; baselines vary widely across patients

These ten characteristics are the design specification. Every subsequent choice in this paper about what to discard, how to smooth, how to construct ground truth, which change-point detectors to consider was guided by these characteristics. Rather than imposing assumptions that

simplify the signal, our approach was designed to respect and preserve the intrinsic properties of the data throughout the analysis pipeline.

3 Data Handling and Preprocessing

3.1 Per-Patient Ingestion and HRV Isolation

Data were received on a per-patient basis. We isolated the HRV channel for cleaning and smoothing while leaving every other modality (steps, SpO₂, skin temperature, heart rate, sleep) untouched at this stage; these covariates are re-introduced onto the common timeline only after HRV has been cleaned and smoothed.

3.2 Exploratory Data Analysis

An exploratory analysis of the isolated HRV signal conducted at both the per-patient level and across the pooled cohort surfaced the ten characteristics enumerated in §2.2. Plotting the value distributions revealed a clear bimodal structure at both scales, with two distinct peaks rather than a single central tendency (Figure 2). This bimodality was treated as a structural property of the cohort and propagated forward as a constraint on every downstream modeling choice.

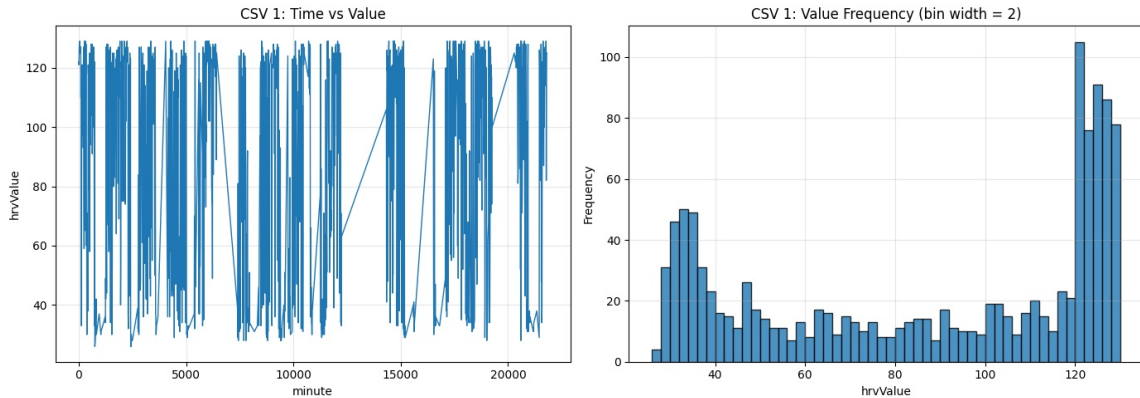


Figure 2: Raw HRV signal (left) and value-frequency histogram (right) for a representative patient, showing the bimodal distribution with two distinct peaks rather than a single central tendency.

3.3 Characterizing Missingness

The EDA also confirmed that missingness in the HRV channel was substantial and of mixed type: both random (transient artifact, brief contact loss) and non-random (device removal, charging, activity-coupled dropouts) gaps were present in every patient’s record. Because the two types carry different physiological meanings, they could not be handled with a single blanket imputation rule.

3.4 The 180-Minute Chunking Rule

HRV in this dataset arrives as a stream of roughly one reading every ten minutes. A continuous stretch is one in which successive readings are separated by approximately this cadence. When a gap appears for example, readings at $t = 10, 20, 30, 100$, then nothing until 300 the question is whether the post-gap segment belongs to the same continuous record as the pre-gap segment, or whether it should be treated as a fresh recording.

In consultation with domain experts, we consider a 180-minute threshold: any inter-reading gap exceeding 180 minutes terminates the current segment, and the next available reading begins a new segment. In the example above, the readings at 10, 20, 30, and 100 are retained as one continuous chunk; the gap from 100 to 300 (200 minutes) exceeds the threshold and triggers a chunk boundary; readings from 300 onward (310, 320, 330, ...) form a new continuous chunk.

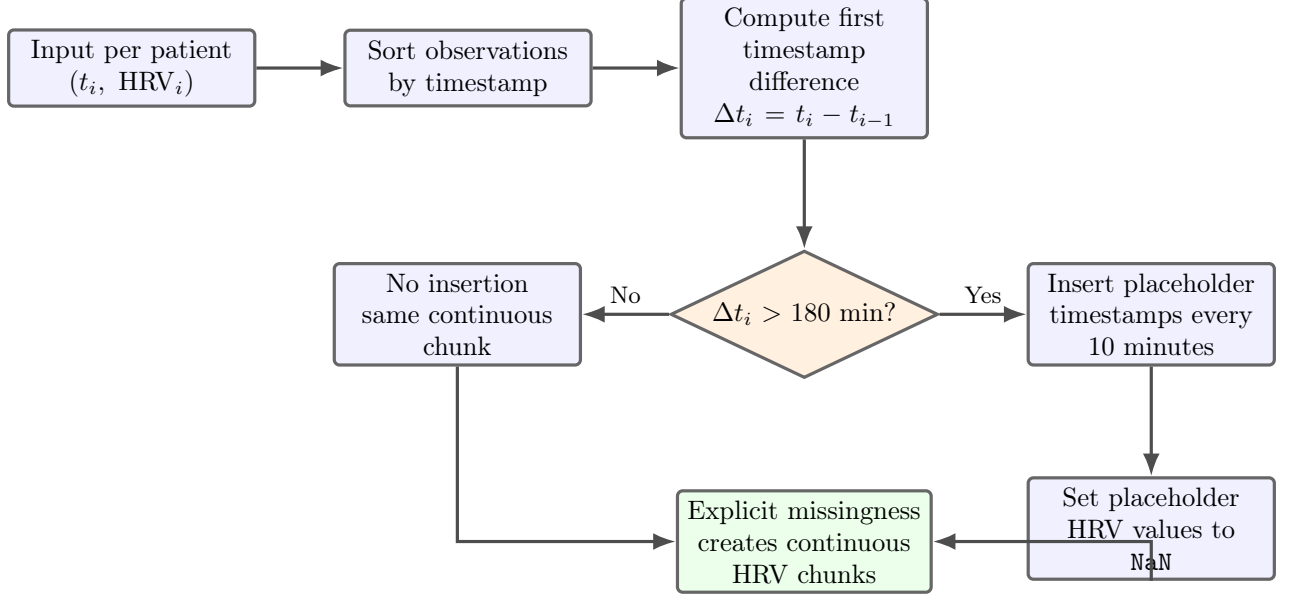


Figure 3: Gap handling before smoothing. HRV observations are sorted by timestamp, first differences between successive timestamps are computed, and only gaps exceeding 180 minutes receive 10-minute NaN placeholder rows. These placeholders expose missingness and naturally define continuous chunks.

The 180-minute cutoff was set by the domain experts on the basis of the empirical distribution of inter-reading intervals locating the elbow points in the gap-duration distribution (median, second, and third modes) combined with physiological judgment about the timescale over which pre-gap autonomic context can be considered informative for post-gap dynamics. Importantly, data within retained chunks are not interpolated or back-filled; the threshold operates only as a segmentation rule, not as an imputation rule, preserving the missingness structure for the pseudo-ground-truth construction in §4.

4 Handling Missingness

Missingness in this dataset is not a nuisance to be removed; it is a structural property of the signal that the pipeline must respect at every stage.

4.1 Why Standard Imputation Fails Here

Mean imputation, forward-fill, and interpolation all assume that the gap is a passive absence of measurement. In wearable HRV the gap is rarely passive: the device was removed, the user was active, contact was poor, or the signal was being lost precisely because something physiologically interesting was happening. Replacing such gaps with smooth interpolants would modify the raw data which leads to a possibility of incorrect predictions.

4.2 Where We Do and Do Not Impute

We do not impute raw HRV for the purpose of producing a cleaner-looking signal. Imputation enters the pipeline only inside the self-supervised pseudo-ground-truth construction (§4), where it serves a different function: a model is trained to predict held-out values, and the prediction error, not the imputed value, is the signal of interest.

5 Smoothing

5.1 Why Smoothing in the First Place

Several of the characteristics identified in §2.2 heteroscedastic noise, artifact contamination, and non-stationarity in particular guarantee that any analysis run directly on raw HRV will be driven by noise rather than by physiology. A smoothing stage that estimates the latent physiological state under noise is therefore a structural requirement before any downstream modeling can proceed.

5.2 Cross-Disciplinary Survey and Scoring System

Rather than default to a familiar smoother, we conducted a structured survey of smoothing and state-estimation methods drawn from across fields like statistics, signal processing, physics, control theory, econometrics, and machine learning. Each method was scored against the ten HRV characteristics (H1–H10) using a simple rubric: 1 point if the method natively handles or makes assumptions consistent with the characteristic, 0.5 points if it can be modified to handle the characteristic with techniques already documented in the literature, and 0 points otherwise. The full survey appears as Table 4 in Appendix A.

5.3 Selection of the Particle Filter

The particle filter (sequential Monte Carlo, SMC) scored highest under this rubric. It natively handles 7 of the 10 characteristics and can be modified to handle the remaining 3 using extensions that are well established in the literature. On this basis we selected the particle filter as the smoothing backbone and proceeded to implement a modified version tailored to HRV; the specific modifications are detailed in §6.

Figure 4 shows the output of the structural particle smoother on a representative patient: the raw HRV stream, the same stream with >180-minute gap markers, the smoothed signal (level + circadian) with its 95% credible band, and the underlying trend (level only, circadian removed).

6 Structural Particle Smoother with Harmonic Components

This section formalizes the particle smoother selected in §5.3 as our HRV smoothing backbone. The mathematical content below specifies the state-space model, the observation model with missing-data handling, the data-driven parameter estimation procedure, and the sequential Monte Carlo forward filter and forward-filter backward-sampler used to produce smoothed trajectories and credible intervals.

6.1 Overview and Motivation

This implementation extends the basic local-level-plus-slope particle smoother to a full *structural time series* model (Harvey, 1989). Three components are superimposed to model HRV:

1. A **local linear trend** (ℓ_t, s_t) capturing the slowly varying physiological baseline.

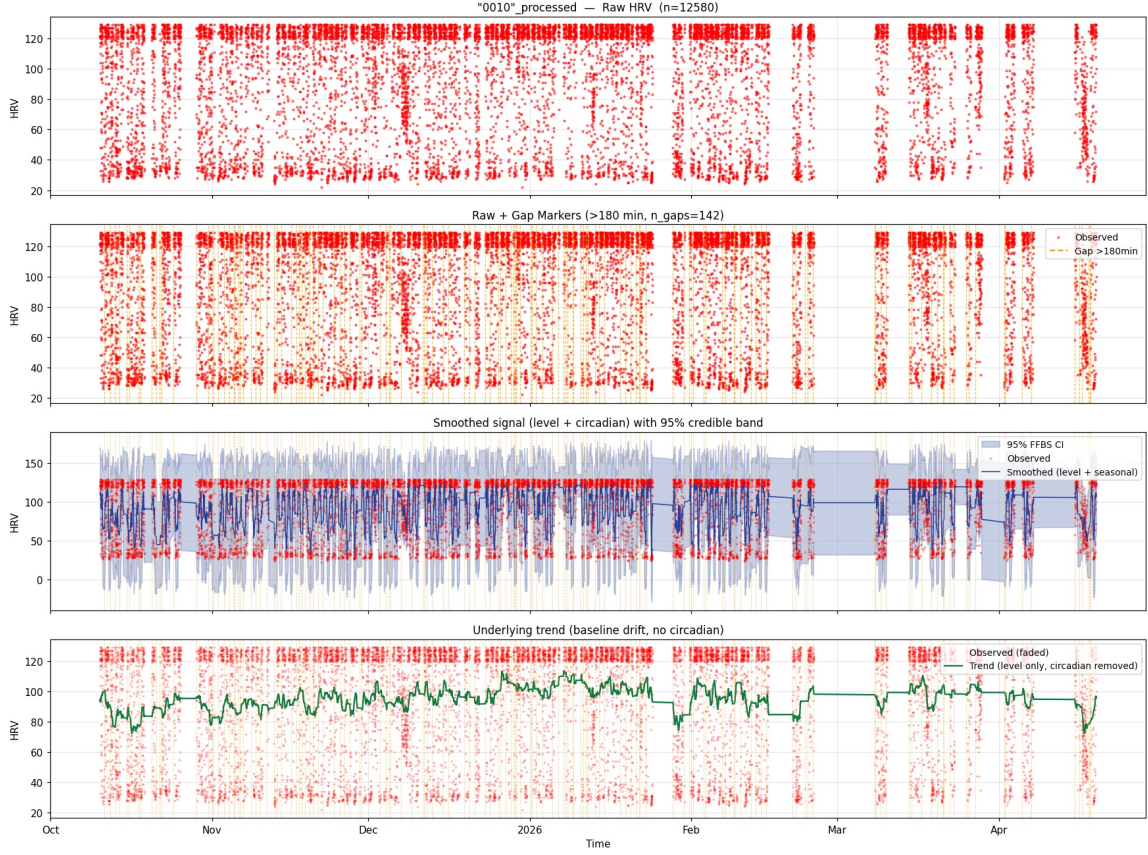


Figure 4: Structural particle smoother outputs on patient 0010. Top to bottom: raw HRV; raw with >180 -minute gap markers; smoothed signal (level + circadian) with 95% credible band; underlying trend (level only, circadian removed).

2. A **daily harmonic** $(c_t^{(1)}, c_t^{*(1)})$ capturing circadian (24-hour) HRV rhythms.
3. A **semi-daily harmonic** $(c_t^{(2)}, c_t^{*(2)})$ capturing the 12-hour ultradian rhythm (rest/activity alternation).

The observation model uses a Student- t likelihood (heavy tails, robust to motion artefacts) and explicitly handles missing observations via a flat likelihood, so the filter does not require pre-segmentation into continuous chunks. Parameters are estimated from the data before filtering using a robust median-absolute-deviation (MAD) procedure.

6.2 State Vector and Initial Distribution

Definition 6.1 (State Vector). At time index $t \in \{1, \dots, T\}$, the latent state is the six-dimensional vector

$$\mathbf{x}_t = (\ell_t, s_t, c_t^{(1)}, c_t^{*(1)}, c_t^{(2)}, c_t^{*(2)})^\top \in \mathbb{R}^6,$$

where ℓ_t is the HRV level, s_t is the slope (rate of change), $(c_t^{(j)}, c_t^{*(j)})$ are the cosine and sine components of the j -th harmonic.

Definition 6.2 (Initial Distribution). The prior at $t = 0$ is the product of independent Gaussians:

$$\begin{aligned}\ell_0 &\sim \mathcal{N}(\mu_\ell^{(0)}, [\sigma_\ell^{(0)}]^2), \\ s_0 &\sim \mathcal{N}(0, 0.09), \\ c_0^{(1)}, c_0^{*(1)} &\sim \mathcal{N}(0, 35^2), \\ c_0^{(2)}, c_0^{*(2)} &\sim \mathcal{N}(0, 15^2),\end{aligned}$$

where $(\mu_\ell^{(0)}, \sigma_\ell^{(0)})$ are estimated from the first day of data (see Section 6.5). The large harmonic variances $(35^2, 15^2)$ encode diffuse prior ignorance about the initial seasonal amplitude.

6.3 State Transition Model

6.3.1 Adaptive Time Increment

Definition 6.3 (Normalised Time Increment). Let $\delta_t^{\text{raw}} := t_t - t_{t-1}$ denote the raw inter-observation gap in minutes. The *normalised time increment* is

$$\Delta_t := \frac{\delta_t^{\text{raw}}}{\tilde{\delta}}, \quad \tilde{\delta} := \text{median}(\{\delta_t^{\text{raw}} : \delta_t^{\text{raw}} > 0\}).$$

At the nominal sampling rate $\tilde{\delta}$, $\Delta_t = 1$. For a double-length gap $\Delta_t = 2$; for a shorter-than-usual step $\Delta_t < 1$. $\Delta_1 := 1$ by convention.

Remark 6.4 (Dimensional Invariance). Since Δ_t is dimensionless, the noise scaling $\sigma\sqrt{\Delta_t}$ depends only on the ratio of the current gap to the median gap, not on the absolute time unit. This makes the model parameters interpretable in units of the typical step regardless of whether the wearable records every 10 or every 20 minutes.

6.3.2 Local Linear Trend Transition

Definition 6.5 (Level and Slope Dynamics). The level and slope evolve as an integrated random walk:

$$\ell_t \mid \ell_{t-1}, s_{t-1} \sim \mathcal{N}(\ell_{t-1} + s_{t-1} \Delta_t, \sigma_\ell^2 \Delta_t), \quad (1)$$

$$s_t \mid s_{t-1} \sim \mathcal{N}(s_{t-1}, \sigma_s^2 \Delta_t). \quad (2)$$

with $\sigma_\ell = 0.05 \hat{\sigma}_{\text{obs}}$ and $\sigma_s = 0.02 \sigma_\ell$.

Proposition 6.6 (Continuous-Time Limit). As $\Delta_t \rightarrow 0$, (1)–(2) become the Itô SDEs $d\ell_t = s_t dt + \sigma_\ell dW_\ell$ and $ds_t = \sigma_s dW_s$, i.e. a second-order (integrated) Wiener process. The discrete model is the exact time-discretisation of this SDE with variance linear in Δ_t (consistent with Brownian scaling).

Proof. For a Brownian motion W and elapsed time Δ_t , $W(\Delta_t) - W(0) \sim \mathcal{N}(0, \Delta_t)$. Multiplying by σ gives the increment $\mathcal{N}(0, \sigma^2 \Delta_t)$. The mean term $s_{t-1} \Delta_t$ is Euler–Maruyama integration of the drift s_t . $\square$

6.3.3 Trigonometric Harmonic Transition

Definition 6.7 (Rotation Matrix). For harmonic $j \in \{1, 2\}$ with angular frequency ω_j , define the rotation by angle $\omega_j \Delta_t$:

$$R_j(\Delta_t) := \begin{pmatrix} \cos(\omega_j \Delta_t) & \sin(\omega_j \Delta_t) \\ -\sin(\omega_j \Delta_t) & \cos(\omega_j \Delta_t) \end{pmatrix}.$$

The harmonic state evolves as

$$\begin{pmatrix} c_t^{(j)} \\ c_t^{*(j)} \end{pmatrix} = R_j(\Delta_t) \begin{pmatrix} c_{t-1}^{(j)} \\ c_{t-1}^{*(j)} \end{pmatrix} + \boldsymbol{\eta}_j, \quad \boldsymbol{\eta}_j \sim \mathcal{N}(\mathbf{0}, \sigma_{\text{seas}}^2 \mathbf{I}), \quad (3)$$

with $\sigma_{\text{seas}} = 0.15$ (fixed).

Theorem 6.8 (Harmonic Correctness). *In the absence of process noise ($\sigma_{\text{seas}} = 0$), if $c_0^{(j)} = A \cos \phi$ and $c_0^{*(j)} = A \sin \phi$ for some amplitude $A \geq 0$ and phase $\phi \in [0, 2\pi)$, then at time t :*

$$c_t^{(j)} = A \cos(\phi - \omega_j t), \quad c_t^{*(j)} = A \sin(\phi - \omega_j t).$$

That is, the rotation propagates a sinusoid of frequency ω_j and constant amplitude A forward in time.

Proof. By induction. At $t = 1$ (and $\Delta_t = 1$ for clarity):

$$\begin{aligned} c_1^{(j)} &= A \cos \phi \cos \omega_j + A \sin \phi \sin \omega_j = A \cos(\phi - \omega_j), \\ c_1^{*(j)} &= -A \cos \phi \sin \omega_j + A \sin \phi \cos \omega_j = A \sin(\phi - \omega_j). \end{aligned}$$

Assume $c_{t-1}^{(j)} = A \cos(\phi - \omega_j(t-1))$, $c_{t-1}^{*(j)} = A \sin(\phi - \omega_j(t-1))$. Applying $R_j(1)$:

$$\begin{aligned} c_t^{(j)} &= A \cos(\phi - \omega_j(t-1)) \cos \omega_j + A \sin(\phi - \omega_j(t-1)) \sin \omega_j = A \cos(\phi - \omega_j t), \\ c_t^{*(j)} &= -A \cos(\phi - \omega_j(t-1)) \sin \omega_j + A \sin(\phi - \omega_j(t-1)) \cos \omega_j = A \sin(\phi - \omega_j t). \end{aligned}$$

The amplitude $\|(c_t^{(j)}, c_t^{*(j)})\| = A$ is preserved because $R_j \in \text{SO}(2)$ is orthogonal ($\det R_j = 1$, $R_j^\top R_j = \mathbf{I}$). $\square$

Corollary 6.9 (Period). *The observable contribution $c_t^{(j)}$ completes one cycle when $\omega_j t = 2\pi$, i.e. after $T_j = 2\pi/\omega_j$ time steps.*

With $\omega_1 = 2\pi/T_{\text{day}}$ (where T_{day} is the number of readings per 24 hours): period = 1 day. With $\omega_2 = 2\pi/(T_{\text{day}}/2)$: period = 12 hours.

6.3.4 Observable Equation

Definition 6.10 (Observable Mean). The mean of the observation at time t is the sum of the three components:

$$\mu_t := \ell_t + c_t^{(1)} + c_t^{(2)}. \quad (4)$$

This decomposes the observed HRV into: slow physiological trend (ℓ_t) plus circadian oscillation ($c_t^{(1)}$) plus ultradian oscillation ($c_t^{(2)}$).

6.4 Observation Model and Missing Data

6.4.1 Student- t Likelihood

Definition 6.11 (Observation Distribution). For a non-missing observation $y_t \in \mathbb{R}$:

$$y_t | \mathbf{x}_t \sim t_\nu(\mu_t, \sigma_{\text{obs}}^2), \quad (5)$$

where $\mu_t = \ell_t + c_t^{(1)} + c_t^{(2)}$, $\nu = 4$, and σ_{obs} is estimated from the data (Section 6.5).

Theorem 6.12 (Robustness of Student- t). *The log-likelihood under the Student- t model grows as $-\frac{\nu+1}{2} \log(1 + e^2/(\nu\sigma^2))$ for residual $e = y_t - \mu_t$ (logarithmic in $|e|$), versus $-e^2/(2\sigma^2)$ under the Gaussian model (quadratic). The ratio of penalties tends to zero as $|e| \rightarrow \infty$, so motion artefact spikes exert negligible relative influence on particle weights compared to a Gaussian likelihood.*

Proof. See the proof of Theorem 2.1 in the standalone `particle_filter.tex` (first version): the ratio $\log(1 + u)/u \rightarrow 0$ as $u \rightarrow \infty$ with $u = e^2/(\nu\sigma^2)$. $\square$

6.4.2 Missing Observation Handling

Definition 6.13 (Flat Likelihood for Missing Data). When $y_t = \text{NaN}$, the observation model is replaced by:

$$p(y_t | \mathbf{x}_t) \propto t_\nu(\mu_t, \sigma_\infty^2), \quad \sigma_\infty = 10^6. \quad (6)$$

Theorem 6.14 (Flat Likelihood Implies Dynamics-Only Update). *Under (6), the importance weights of any two particles $\mathbf{x}_t^{(i)}$ and $\mathbf{x}_t^{(j)}$ satisfy*

$$\frac{w_t^{(i)}}{w_t^{(j)}} = \left(\frac{1 + (y_t - \mu_t^{(i)})^2 / (\nu \sigma_\infty^2)}{1 + (y_t - \mu_t^{(j)})^2 / (\nu \sigma_\infty^2)} \right)^{-(\nu+1)/2} \rightarrow 1 \quad \text{as } \sigma_\infty \rightarrow \infty,$$

for any bounded $\mu_t^{(i)}, \mu_t^{(j)}$. Consequently all particles receive equal weight, the observation is ignored, and the filter propagates solely on the transition dynamics (1)–(3).

Proof. With $\sigma_\infty = 10^6$, the denominator $\nu \sigma_\infty^2 = 4 \times 10^{12}$. For any bounded $|y_t - \mu_t^{(i)}|$:

$$\frac{(y_t - \mu_t^{(i)})^2}{\nu \sigma_\infty^2} \leq \frac{M^2}{4 \times 10^{12}} \approx 0.$$

Both numerator and denominator fractions $\rightarrow 1$, so the ratio $\rightarrow 1$. In practice for HRV values bounded in $[0, 300]$ ms, the weight ratio deviates from 1 by at most $\mathcal{O}(10^{-8})$. $\square$

Remark 6.15 (Superiority over Hard Chunking). Theorem 6.14 shows that the flat likelihood achieves gap-handling *within the single filter pass* without segmenting the time series into independent chunks. The harmonic and trend states continue to evolve through the gap according to their dynamics, so the filter correctly extrapolates the trend across a missing period and resumes tracking when observations resume. Hard chunking (used by other smoothers in this pipeline) discards all cross-gap information; the flat-likelihood approach retains it.

6.5 Data-Driven Parameter Estimation

Definition 6.16 (MAD Estimator of Observation Noise). Let $\Delta y_i = y_i - y_{i-1}$ denote the first-difference sequence with NaN values removed. The observation noise standard deviation is estimated as:

$$\hat{\sigma}_{\text{obs}} = \max \left(\frac{1.4826}{\sqrt{2}} \text{MAD}(\Delta y), 3.0 \right),$$

where $\text{MAD}(\Delta y) = \text{median}|\Delta y_i - \text{median}(\Delta y)|$.

Theorem 6.17 (Consistency of the MAD Estimator). *For i.i.d. $\varepsilon_t \sim \mathcal{N}(0, \sigma^2)$ and a slowly varying trend such that $|f_t - f_{t-1}| \ll |\varepsilon_t - \varepsilon_{t-1}|$:*

$$\frac{1.4826}{\sqrt{2}} \text{MAD}(\Delta y) \xrightarrow{T \rightarrow \infty} \sigma \quad \text{in probability.}$$

Proof. Step 1: Under the approximation $\Delta y_t \approx \varepsilon_t - \varepsilon_{t-1}$, $\Delta y_t \sim \mathcal{N}(0, 2\sigma^2)$ i.i.d. (since $\varepsilon_t, \varepsilon_{t-1}$ are independent). Therefore $\Delta y_t / (\sigma\sqrt{2}) \sim \mathcal{N}(0, 1)$.

Step 2: For $Z \sim \mathcal{N}(0, 1)$, the population MAD is $\text{med}|Z - \text{med}(Z)| = \text{med}|Z|$ (since $\text{med}(Z) = 0$). Now $P(|Z| \leq c) = 2\Phi(c) - 1$, so $\text{med}|Z| = \Phi^{-1}(3/4) \approx 0.6745$. Thus $\text{MAD}(Z) = 0.6745$ and $Z = \text{MAD}(Z)/0.6745 = 1.4826 \text{MAD}(Z)$.

Step 3: Since $\Delta y_t = \sigma\sqrt{2} Z_t$: $\text{MAD}(\Delta y) = \sigma\sqrt{2} \text{MAD}(Z) = 0.6745\sigma\sqrt{2}$, and $\frac{1.4826}{\sqrt{2}} \text{MAD}(\Delta y) = \frac{1.4826}{\sqrt{2}} \cdot 0.6745\sigma\sqrt{2} = 1.4826 \times 0.6745\sigma \approx \sigma$. Consistency of the sample MAD follows from the consistency of sample quantiles. $\square$

Definition 6.18 (Initial Level Estimation). Let $n_{\text{day}} = \max(\lfloor 1440/\tilde{\delta} \rfloor, 50)$ be the approximate number of readings in the first 24 hours. The initial parameters are:

$$\mu_{\ell}^{(0)} = \text{median}(y_{1:n_{\text{day}}}), \quad \sigma_{\ell}^{(0)} = \max(2 \text{ MAD}(y_{1:n_{\text{day}}}), 10).$$

If fewer than 5 non-missing values are available, the global median and $\sigma_{\ell}^{(0)} = 20$ are used instead.

Definition 6.19 (Angular Frequencies).

$$\omega_1 = \frac{2\pi}{T_{\text{day}}}, \quad \omega_2 = \frac{2\pi}{T_{\text{day}}/2} = \frac{4\pi}{T_{\text{day}}}, \quad T_{\text{day}} := \frac{1440}{\tilde{\delta}}.$$

These are data-driven: if $\tilde{\delta} = 10$ min then $T_{\text{day}} = 144$, $\omega_1 = 2\pi/144 \approx 0.0436$ rad/step; if $\tilde{\delta} = 20$ min then $T_{\text{day}} = 72$, $\omega_1 = 2\pi/72 \approx 0.0873$ rad/step.

Definition 6.20 (Derived Noise Scales).

$$\sigma_{\ell} = 0.05 \hat{\sigma}_{\text{obs}}, \quad \sigma_s = 0.02 \sigma_{\ell} = 0.001 \hat{\sigma}_{\text{obs}}, \quad \sigma_{\text{seas}} = 0.15.$$

The hierarchy $\sigma_s \ll \sigma_{\ell} \ll \sigma_{\text{obs}}$ encodes the prior that the trend changes slowly relative to noise, and the slope changes even more slowly than the level.

6.6 Sequential Monte Carlo Forward Filter

Definition 6.21 (Bootstrap Particle Filter). Approximate $p(\mathbf{x}_t | y_{1:t})$ by $\sum_{i=1}^N w_t^{(i)} \delta_{\mathbf{x}_t^{(i)}}$ with N particles. At each step t :

1. **Propagate:** $\tilde{\mathbf{x}}_t^{(i)} \sim p(\mathbf{x}_t | \mathbf{x}_{t-1}^{(i)})$ via (1)–(3).
2. **Weight:** $\tilde{w}_t^{(i)} \propto p(y_t | \tilde{\mathbf{x}}_t^{(i)})$ via (5) or (6).
3. **Normalise:** $w_t^{(i)} = \tilde{w}_t^{(i)} / \sum_j \tilde{w}_t^{(j)}$.
4. **Systematic resampling:** draw N indices with probabilities $\{w_t^{(i)}\}$; reset weights to $1/N$.

Parameters: $N = 100$ particles (default; set per patient from Phase 1 results).

Theorem 6.22 (Filter Consistency). *For any bounded continuous φ , $\sum_i w_t^{(i)} \varphi(\mathbf{x}_t^{(i)}) \rightarrow \mathbb{E}[\varphi(\mathbf{x}_t) | y_{1:t}]$ a.s. as $N \rightarrow \infty$.*

Proof. Del Moral (2004), Theorem 7.4.4. The transition density is non-degenerate (Gaussian) and the importance weights are bounded because the Student- t observation density is bounded above for any fixed y_t . $\square$

6.7 Forward-Filter Backward-Sampler

Definition 6.23 (FFBS Algorithm). Draw M approximate samples from the joint smoothing distribution $p(\mathbf{x}_{1:T} | y_{1:T})$ using the stored filter history:

1. Run the bootstrap filter; store $\{\mathbf{x}_t^{(i)}, w_t^{(i)}\}_{i,t}$.
2. For $m = 1, \dots, M$: draw $\mathbf{x}_T^{(m)}$ from the terminal filter; then for $t = T-1, \dots, 1$ sample $\mathbf{x}_t^{(m)}$ with probability proportional to $w_t^{(i)} \cdot p(\mathbf{x}_{t+1}^{(m)} | \mathbf{x}_t^{(i)})$.

Parameters: $M = 50$ backward paths (default).

Definition 6.24 (Smoothed Outputs). From the M backward trajectories, four quantities are computed at each t :

$$\begin{aligned}\text{smoothed_hrv}_t &= \frac{1}{M} \sum_{m=1}^M (\ell_t^{(m)} + c_t^{(1,m)} + c_t^{(2,m)}), \\ \text{true_trend_level}_t &= \frac{1}{M} \sum_{m=1}^M \ell_t^{(m)}, \\ \text{ci_lower_95}_t &= P_{2.5} \left(\ell_t^{(m)} + c_t^{(1,m)} + c_t^{(2,m)} \right), \\ \text{ci_upper_95}_t &= P_{97.5} \left(\ell_t^{(m)} + c_t^{(1,m)} + c_t^{(2,m)} \right),\end{aligned}$$

where $P_q(\cdot)$ denotes the q -th sample percentile over $m = 1, \dots, M$.

Remark 6.25 (Decomposition). The `smoothed_hrv` output recovers the full observable signal $\mu_t = \ell_t + c_t^{(1)} + c_t^{(2)}$, which includes all variability that is physiologically structured. The `true_trend_level` isolates the slow baseline by removing the harmonic components, providing a deseasonalised HRV trend that is more sensitive to gradual physiological changes.

6.8 Assumptions of the Smoothing Stage

Assumption 6.26. HRV decomposes additively into a slow trend (ℓ_t), a circadian oscillation ($c_t^{(1)}$), and a semi-daily oscillation ($c_t^{(2)}$), plus heavy-tailed noise.

Assumption 6.27. The median inter-observation gap $\tilde{\delta}$ is a stable representative of the typical sampling rate. Sessions with highly variable sampling rates may require a different normalisation.

Assumption 6.28. Observation noise is i.i.d. Student- t with $\nu = 4$ degrees of freedom. This is fixed across all patients; a patient-specific ν could be estimated but is not implemented.

Assumption 6.29. $N = 100$ particles provides a practically adequate approximation for the series lengths encountered in the AIIMS dataset. Long series (several months) may require larger N to prevent particle degeneracy.

7 Change-Point Detection

With a smoothed signal in hand and missingness explicitly accounted for, we now turn to the second main objective of this pipeline: detecting changes in the physiological signal before any machine learning model is brought in. What we have at this stage is a trend line that is actionable: an expert clinician inspecting the smoothed trace can point to regions where the state appears to shift, where the trend bends, or where the underlying dynamics look qualitatively different from the surrounding context. This is, in essence, a manual annotation task; our objective is to automate it.

Change-point detection (CPD) is the natural framing for this problem. The technique is not native to clinical physiology: it has a long history in fault detection, finance, control systems, engine and system failure monitoring, chemistry, physics, astronomy, and aerospace, where the common need is to flag the abrupt or abnormal in a data stream so that downstream decisions can be made on top of it. We adopt CPD here in exactly that spirit.

It is important to be precise about what change-point detection does and does not provide: a detected change point is a *marker* that something in the data has changed; it is not, by itself, a physiological claim.

These CPD methods are purely mathematical in nature. Because of this, they can flag even very

small changes in the signal, many of which may not be physiologically meaningful. Therefore, while the methods can identify where changes occur, deciding which of those changes are actually relevant from a physiological perspective requires domain expertise. In practice, the **thresholds and criteria** used to interpret the outputs of these methods should be determined by clinicians or other experts with a great understanding of the underlying physiology.

Interpretation of the change in physiological terms is left to a downstream stage, ultimately the machine learning model. The role of CPD in our pipeline is therefore to *tag* the **smoothed HRV** trace with change-point annotations before that trace is handed off to the model.

Selecting a CPD method on a signal with the properties enumerated in §2.2 is not a free choice. As in the smoothing stage, we cannot apply an arbitrary off-the-shelf detector and trust the result; the ten characteristics H1–H10 act as binding constraints. We therefore followed the same approach we used for smoothing: we surveyed change-point detection methods across the fields in which they originated finance, chemistry, physics, engineering, engine and system failure analysis, astronomy, and related areas to build an arsenal of candidate methods from which a principled selection can be made against H1–H10. The full survey is extensive; rather than reproduce it inline, we provide a link to the complete document at the end of this section. What follows in §8 presents the subset of methods we plan to implement on the smoothed HRV signal, together with the change types each is primarily suited to detect.

8 Candidate Change-Point Detection Methods

We plan to implement four candidate change-point detection (CPD) methods on the smoothed HRV signal: Bayesian Online Change-Point Detection (BOCPD), Kernel Change-Point Detection (KCP), the Kalman filter, and Hidden Markov Models (HMM). Each method is primarily suited to a different class of change. Table 3 summarises the coverage of each method across the relevant change types and shows that, taken together, the four methods collectively cover all change types of interest.

Table 3: Coverage of candidate change-point detection methods across change types. Together, the four methods cover all required change types. [Highlighted cell](#) denotes the method for which that change type is its primary strength.

Change Type	BOCPD	KCP	Kalman	HMM
Mean shift	✓ Primary	✓ Detects	✓ Via innovations	✓ Via state
Variance change	✓ Primary	✓ Detects	✓ Via EWMA	~ Partial
Distributional shape	✗ Blind	✓ Primary	✗ Blind	✗ Blind
Gradual drift / trend	✗ Late	✗ Late	✓ Primary	✗ Late
Autocorrelation change	✗ Blind	✗ Blind	✓ Primary	✗ Blind
Multivariate joint	✗ Univariate	✓ Primary	✗ Univariate	~ Partial
Recurring states	✓ Via resets	✗ No states	✗ Continuous	✓ Primary
Missing data / gaps	~ Via epochs	~ Via epochs	✓ Native	~ Special handling

A complete survey of candidate change-point detection methods (including those not selected for implementation) is available as a separate document:

This is our change-point detection survey.

9 Mathematical Formulation of Selected CPD Methods

Each of the four methods selected in §3.3 — BOCPD, Kalman filter, KCP, and HMM — is developed below with its formal definition, the parameters we instantiate for the HRV setting, and the derivation underpinning its change-detection behavior. Empirical results on the smoothed HRV signal accompany each method, with a consolidated overlay at the end of this section.

9.1 Bayesian Online Change-Point Detection (BOCPD)

[Placeholder for derivation, parameter choices, and proof of correctness for BOCPD as applied to the smoothed HRV residual signal.]

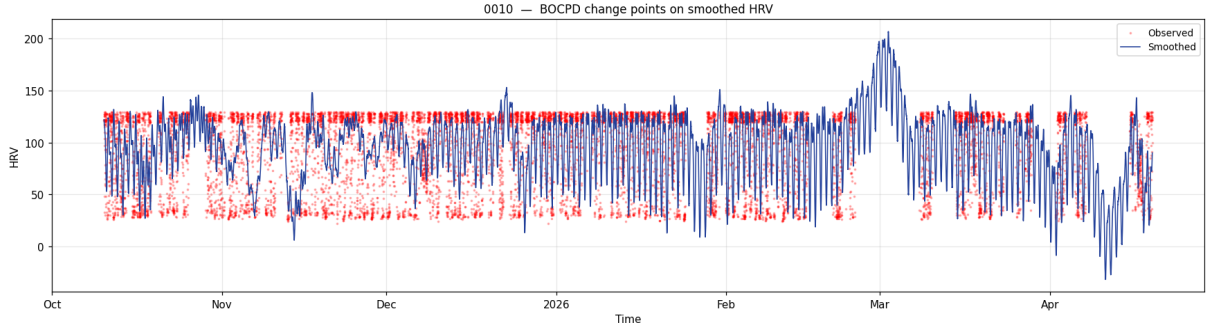


Figure 5: BOCPD annotations overlaid on smoothed HRV.

9.2 Kalman Filter for Change-Point Detection

[Placeholder for state-space formulation, innovation sequence derivation, and parameter choices for the Kalman-based detector.]

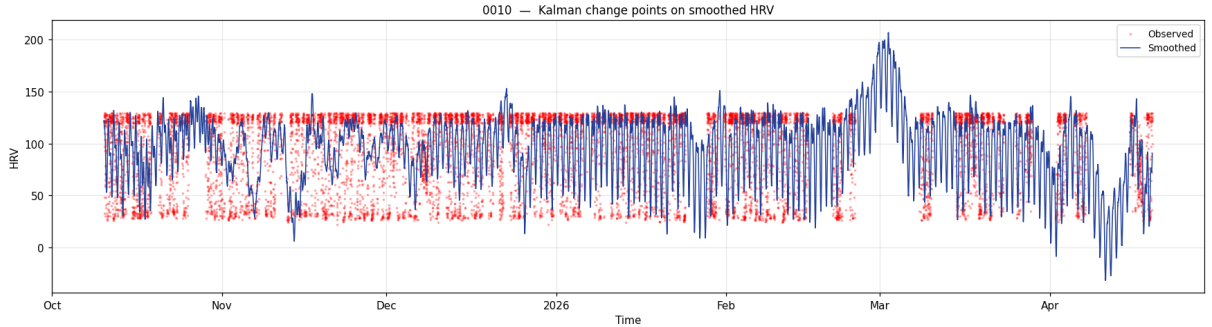


Figure 6: Kalman annotations overlaid on smoothed HRV.

9.3 Kernel Change-Point Detection (KCP)

[Placeholder for kernel choice, MMD-based test statistic, and parameter choices for KCP applied to the smoothed HRV signal.]

9.4 Hidden Markov Models (HMM)

[Placeholder for state definitions, emission and transition models, and inference procedure for the HMM-based detector.]

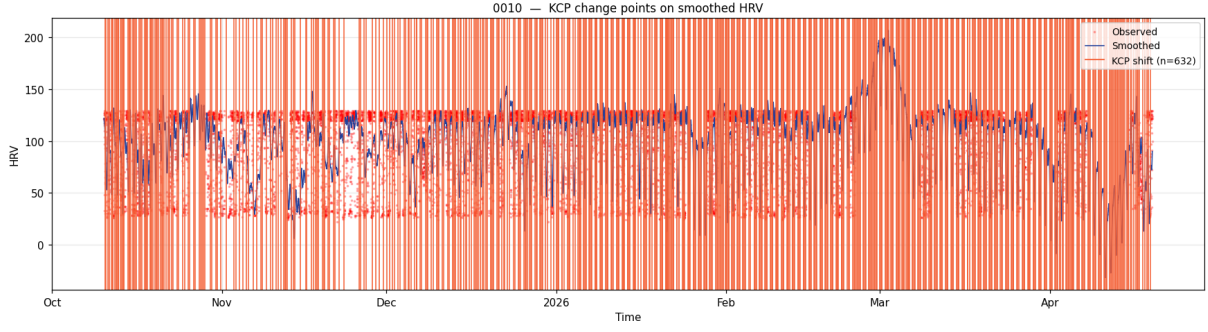


Figure 7: KCP annotations overlaid on smoothed HRV.

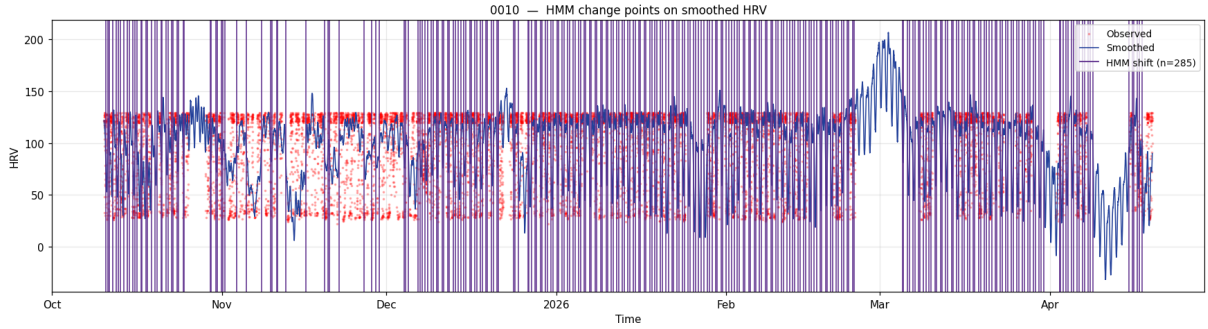


Figure 8: HMM annotations overlaid on smoothed HRV.

9.5 Consolidated Overlay

A consolidated visualization summarizes the behavior of all four methods on a representative segment. The figure is constructed in layered fashion: the **raw HRV signal** forms the base, with **gaps** marked from the missingness mask of §3.2; the **smoothed trend** from the particle filter is overlaid on top; and on this smoothed trend, the **change-point annotations** from BOCPD, Kalman, KCP, and HMM are drawn as separate marker tracks, followed by their union as a final track.

The manuscript concludes here with the implementation of the four candidate change-point detection methods. All four methods: BOCPD, Kalman-based CPD, KCP, and HMM have been implemented and are theoretically and mathematically correct, and have been successfully applied to our smoothed HRV data. However, the detection thresholds governing the transition from **normal** to **transition** to **shift** states are *not yet clinically tuned*. The present thresholds were selected arbitrarily and serve as functional defaults rather than physiologically validated operating points.

As a direct consequence, the methods currently flag a large number of change points in our visualisations; this is an artefact of untuned thresholds rather than a reflection of the underlying signal, and we expect the detection rate to converge to a clinically meaningful regime once the thresholds are calibrated. Calibration requires input from clinicians, physicians, or physiology experts, who are best positioned to define what constitutes a clinically meaningful autonomic shift in this cohort. Such expert-driven threshold tuning, together with downstream experimentation, evaluation, and integration into a deployment pipeline, is out of scope for this manuscript and will be addressed in subsequent work.

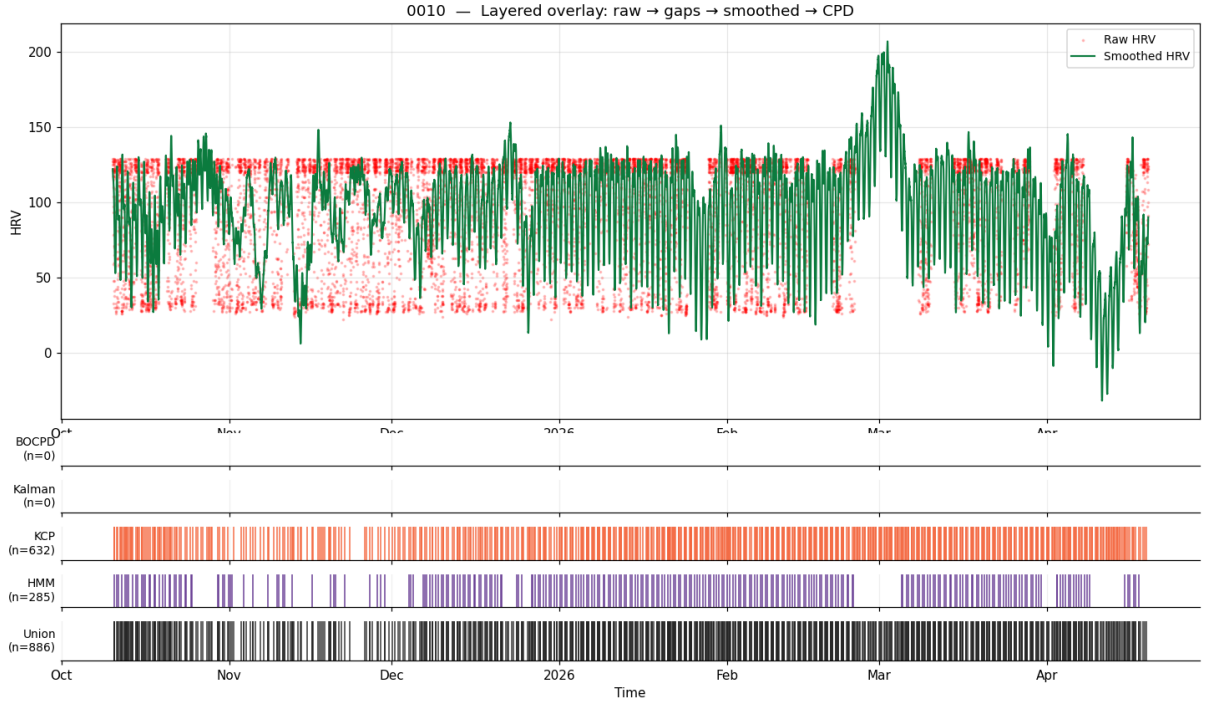


Figure 9: Full overlay: raw HRV → gaps → smoothed trend → BOCPD annotations → Kalman annotations → KCP annotations → HMM annotations → all four combined.

Acknowledgements

We sincerely thank our study partner, GOQII, for providing the multimodal wearable recordings that made this work possible. We are deeply grateful to **Dr. Mayank Garg**, **Dr. Pradeep Tiwari**, and **Dr. Naveen Bhatruja** for their invaluable guidance, mentorship, and continuous support throughout this study. Their clinical expertise and insights played a crucial role in shaping the segmentation strategy and parameter-selection decisions described in this work. We acknowledge that the progress achieved thus far has been possible because of their encouragement, direction, and mentorship at every stage of the project.

Code and Data Availability

All code, scripts, and supporting material presented in this work are available in our GitHub repository:

<https://github.com/jatindangi1206/asthma-prediction.git>

This repository contains the complete implementation of the preprocessing pipeline, the structural particle smoother.

References

References

- [1] Harvey, A.C. (1989). *Forecasting, Structural Time Series Models and the Kalman Filter*. Cambridge University Press.

- [2] Del Moral, P. (2004). *Feynman–Kac Formulae: Genealogical and Interacting Particle Systems with Applications*. Springer.
- [3] Godsill, S., Doucet, A., & West, M. (2004). Monte Carlo Smoothing for Nonlinear Time Series. *Journal of the American Statistical Association*, 99(465), 156–168.
- [4] Kitagawa, G. (1996). Monte Carlo Filter and Smoother for Non-Gaussian Nonlinear State-Space Models. *Journal of Computational and Graphical Statistics*, 5(1), 1–25.

A Smoothing Method Survey

This appendix presents the full survey and scoring used to evaluate smoothing methods against the ten HRV characteristics (H1–H10) and to arrive at the final selection of the particle filter.

Legend. • = method natively handles the characteristic (1.0 point); ◦ = method can be modified to handle the characteristic (0.5 point); · = method does not address the characteristic (0 points); ✕ = method fails on the characteristic. **Surv** = survives filtering; **Cov** = total coverage score.

Table 4: Filtering matrix: 85 smoothing methods $\times$ 10 HRV characteristics. Sorted: survivors first (by coverage descending), then eliminated by family.

Method	Family	H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Surv	Cov
Particle Filter	Kalman	◦	•	◦	•	◦	•	•	•	•	•	✓	8.5
Mixture Kalman Filter	Kalman	◦	•	◦	•	·	•	•	•	•	◦	✓	7.5
LSTM-MDN / GRU-MDN / MDN-RNN	Mixture / HMM	◦	•	•	•	◦	•	◦	◦	◦	•	✓	7.5
Structural TS + Student-t	Kalman	·	·	◦	•	•	•	•	•	•	◦	✓	7.0
SLDS / Switching Kalman	Kalman	◦	•	◦	•	·	◦	•	•	•	◦	✓	7.0
VRNN	Neural VAE	/	◦	•	•	◦	◦	◦	◦	◦	•	✓	7.0
Conditional VAE (CVAE)	Neural VAE	/	◦	•	◦	•	◦	◦	◦	◦	•	✓	7.0
Neural ODE / Latent ODE	Neural VAE	/	·	◦	•	•	◦	◦	•	•	◦	✓	7.0
State-Space GP	Gaussian Process	·	·	◦	◦	◦	•	•	•	•	•	✓	6.5
Bayesian Structural TS	Gaussian Process	·	·	◦	•	•	◦	•	•	◦	•	✓	6.5
TCN-VAE	Neural VAE	/	◦	◦	•	◦	◦	◦	◦	◦	•	✓	6.5
Transformer-VAE	Neural VAE	/	◦	◦	•	◦	◦	•	•	◦	·	✓	6.5
Time-VAE	Neural VAE	/	◦	◦	◦	•	•	◦	◦	◦	◦	✓	6.5
TBATS	Seasonal-Trend	•	◦	◦	•	•	◦	◦	◦	·	◦	✓	6.0
LSTM-VAE	Neural VAE	/	◦	◦	•	◦	◦	◦	◦	◦	•	✓	6.0
BRITS	Neural VAE	/	·	·	•	◦	◦	◦	•	•	◦	✓	6.0
Mixture Density Network (MDN)	Mixture HMM	/	◦	•	◦	◦	◦	•	◦	◦	·	✓	6.0
Skewed-t MDN (Dumitrescu 2026)	Mixture HMM	/	◦	•	◦	◦	◦	•	◦	◦	·	✓	6.0
Lipponen–Tarvainen	Preprocessors	•	◦	◦	·	·	•	•	◦	·	•	✓	5.5
Kubios Cubic-Spline	Preprocessors	•	◦	◦	·	·	•	•	◦	·	•	✓	5.5
Huber-Whittaker	Penalised	·	·	◦	•	◦	•	•	•	·	◦	✓	5.5

continued on next page

Table 4 – continued from previous page

Method	Family	H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Surv	Cov
Robust Kalman (Student-t)	Kalman	•	•	○	○	•	●	●	●	●	○	✓	5.5
Diffusion (DDPM)	Neural VAE	/	○	○	○	○	●	○	○	•	●	✓	5.5
BELMM (Bayesian latent mixture)	Mixture HMM	/	○	●	○	○	○	○	○	•	●	✓	5.5
Hidden Markov Model (Gaussian emit.)	Mixture HMM	/	○	●	○	●	•	○	○	•	●	✓	5.5
Mixture-of-Experts smoothing	Mixture HMM	/	○	●	○	○	○	○	○	•	●	✓	5.5
Gaussian Mixture VAE (GM-VAE / VaDE)	Mixture HMM	/	○	●	○	○	○	○	○	•	●	✓	5.5
Sparse Variational GP	Gaussian Process	•	•	○	○	○	○	●	•	●	●	✓	5.0
DWT Denoising	Wavelet	•	•	●	●	○	●	○	○	•	○	✓	5.0
Stationary Wavelet (SWT)	Wavelet	•	•	●	●	○	●	○	○	•	○	✓	5.0
Empirical Wavelet (EWT)	Wavelet	•	•	●	●	●	○	○	○	•	○	✓	5.0
Wavelet Packet	Wavelet	•	•	●	●	○	●	○	○	•	○	✓	5.0
EEMD / CEEMDAN	Empirical Decomp.	•	•	○	●	●	●	○	○	•	○	✓	5.0
Singular Spectrum (SSA)	Empirical Decomp.	•	•	●	●	●	○	○	○	•	○	✓	5.0
STL (robust)	Seasonal-Trend	•	•	○	●	●	●	○	○	•	○	✓	5.0
MSTL	Seasonal-Trend	•	•	○	●	●	●	○	○	•	○	✓	5.0
Prophet	Seasonal-Trend	•	•	○	●	●	○	○	●	•	○	✓	5.0
CNN-VAE [USER VALIDATED]	Neural VAE	/	○	○	○	○	○	○	○	•	○	✓	5.0
Adaptive Thresh + Cubic Spline	HRV-specific	●	•	○	○	•	●	○	○	•	●	✓	5.0
GP Periodic $\times$ Matérn	Gaussian Process	•	•	○	○	●	○	●	•	•	●	✓	4.5
Lomb–Scargle	FFT	•	•	○	○	●	○	●	○	•	○	✓	4.5
EMD	Empirical Decomp.	•	•	○	●	●	○	○	○	•	○	✓	4.5
Variational Mode Decomp.	Empirical Decomp.	•	•	○	●	●	○	○	○	•	○	✓	4.5
VAE	Neural VAE	/	○	○	○	○	○	○	○	•	•	✓	4.5
β -VAE	Neural VAE	/	○	○	○	○	○	○	○	•	•	✓	4.5
Disentangled VAE (Factor-VAE)	Neural VAE	/	○	○	○	○	○	○	○	•	•	✓	4.5
Savitzky–Golay	Local poly	•	•	○	○	○	○	○	•	●	○	✓	4.0
Robust LOESS	Local poly	•	•	○	●	○	●	○	•	•	○	✓	4.0
P-Splines	Splines	•	•	○	○	○	○	●	○	•	○	✓	4.0
Adaptive Trend Filter	Penalised	•	•	○	●	○	○	○	○	•	○	✓	4.0
GP Student-t likelihood	Gaussian Process	•	•	○	○	•	●	●	•	•	●	✓	4.0
STFT / Spectrogram	FFT	•	•	○	○	●	○	○	○	•	○	✓	4.0
Hilbert–Huang Transform	FFT	•	•	○	●	●	○	○	•	•	○	✓	4.0
Berntson Box-Whisker	Preprocessors	●	○	•	•	•	●	•	•	•	●	✓	3.5
Denoising AE (DAE)	Neural VAE	/	•	•	○	○	○	●	○	•	○	✓	3.5
1D U-Net Denoiser	Neural VAE	/	•	•	○	○	○	○	○	•	•	✓	3.5

continued on next page

Table 4 – continued from previous page

Method	Family	H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Surv	Cov
Self-Supervised / SimCLR	Neural VAE /	·	·	○	○	○	○	○	·	·	●	✓	3.5
Gaussian Mixture Model (Costa 2012)	Mixture HMM /	○	●	○	·	·	·	○	·	·	●	✓	3.5
Smoothness-Priors (Tarvainen)	HRV-specific	●	·	○	○	·	○	·	·	·	●	✓	3.5
Rolling Median	Preprocessors	·	·	○	·	·	●	○	·	·	●	✓	3.0
TVD (ROF)	Penalised	·	·	○	○	·	○	○	○	·	○	✓	3.0
L1 Trend Filtering	Penalised	·	·	○	○	·	○	○	○	·	○	✓	3.0
Vanilla AE	Neural VAE /	·	·	○	○	○	○	○	·	·	○	✓	3.0
Resampling + Spline	HRV-specific	·	·	·	·	·	○	●	○	·	●	✓	3.0
Hampel Filter	Preprocessors	·	·	○	·	·	●	·	·	·	●	✓	2.5
Malik / Task Force 20%	Preprocessors	●	·	·	·	·	○	·	·	·	●	✓	2.5
Simple Moving Average	Classical MA	·	·	○	·	✗	✗	·	✗	●	○	✗	2.0
EWMA	Classical MA	·	·	○	·	✗	✗	·	✗	●	○	✗	2.0
Triangular MA	Classical MA	·	·	○	·	✗	✗	·	✗	●	○	✗	2.0
LOWESS (vanilla)	Local poly	·	·	○	○	✗	✗	○	·	○	○	✗	2.5
Nadaraya–Watson	Local poly	·	·	○	○	✗	✗	●	·	·	○	✗	2.5
Local Polynomial Reg.	Local poly	·	·	○	○	✗	✗	●	·	·	○	✗	2.5
Cubic Smoothing Spline	Splines	·	·	○	○	✗	✗	●	·	·	○	✗	2.5
Whittaker–Eilers	Splines	·	·	○	○	✗	○	●	●	·	○	✗	4.0
Thin-Plate Splines	Splines	·	·	○	○	✗	✗	○	·	·	○	✗	2.0
Hodrick–Prescott	Penalised	·	·	○	✗	✗	✗	·	·	·	○	✗	1.0
Linear Kalman Filter	Kalman	·	·	○	○	✗	✗	●	●	●	○	✗	4.5
RTS Smoother	Kalman	·	·	○	○	✗	✗	●	●	·	○	✗	3.5
UKF (Unscented)	Kalman	·	·	○	○	✗	✗	●	●	·	○	✗	3.5
GPR (RBF kernel)	Gaussian Process	·	·	✗	✗	✗	✗	●	·	·	●	✗	2.0
FFT Low-Pass	FFT	·	·	✗	✗	✗	✗	✗	✗	·	○	✗	0.5
FFT Band-Pass	FFT	·	·	✗	✗	○	✗	✗	✗	·	○	✗	1.0
Wiener Filter	FFT	·	·	✗	✗	✗	✗	✗	✗	·	○	✗	0.5
Welch Periodogram	FFT	·	·	○	○	○	○	✗	✗	·	○	✗	2.5
Multi-Taper Spectral	FFT	·	·	○	○	○	○	✗	✗	·	○	✗	2.5